



Microscope Instructions

Visual inspection and implant characterization are done using a digital microscope provided by BMT.

1. Materials

1. Digital microscope¹
 - a. LCD screen with zoom ring
 - b. Microscope stand
 - c. 2x USB-A to USB-C cable
 - d. AC Adapter
 - e. Mouse
2. Microscope slide with a micrometer scale
3. Implant holder, if needed

2. Assembling the microscope

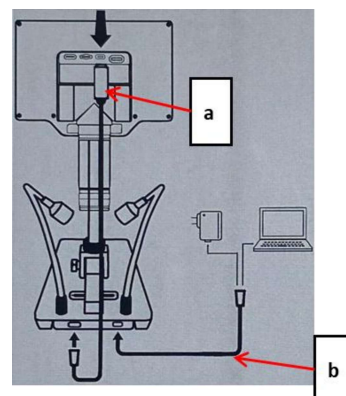
1. Insert the LCD screen into the holder of the microscope stand and hand-tighten the two fixing screws.

Make sure the LCD screen is inserted completely, otherwise the screws will tighten the zoom ring.

3. Setting up power supply

1. Connect the USB-C port of the LCD screen to the USB-A port to the microscope stand (a)
2. Connect the USB-C port of the microscope stand to the AC adapter (b)
3. Connect the AC adapter to the power plug
4. Connect the mouse to the USB-A port of the LCD screen

The microscope should now be supplied with power.

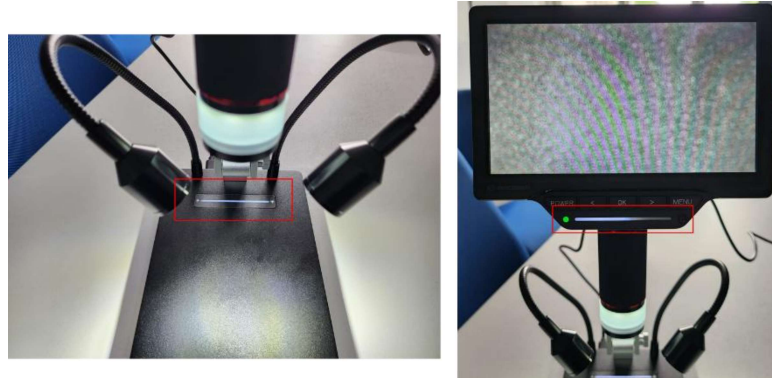


¹ Bresser WiFi 1080P Digital Microscope 2L with LCD (Bresser GmbH, Rhede)

3.1 Adjusting the light

The microscope is equipped with an LED ring light for optimal object illumination. In addition, there are two gooseneck LEDs on the microscope stand.

1. Slide your finger on the touch area on the screen to adjust the light of the LED ring. Or press on the right end for the maximal brightness (left)
2. Slide your finger on the touch area on the stand to adjust the light of the gooseneck LEDs (right).



3.2 Focusing

Coarse focusing is done by moving the camera up and down with the side knobs. Fine focusing is done by using the zoom ring.



3.3 Taking pictures

1. Press the MENU button
2. Use the right arrow button to switch to Settings (forth icon from the left) or use the mouse

If *Foto* is not marked in green:

3. Press OK. The option *Foto/Video* is marked in red.
4. Press OK.
5. Select *Foto*
6. Press OK. *Foto* turns green
7. Press the MENU button to exit the menu.

8. Make sure the SD card is inserted.
9. Press OK to take a picture.

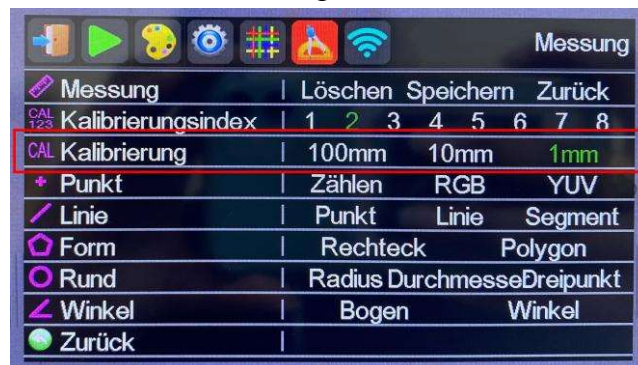


3.4 Measuring

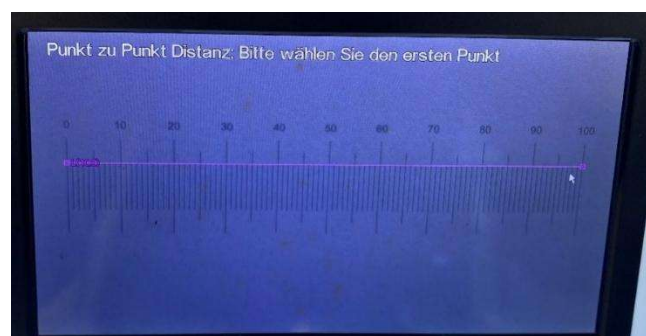
To use the measuring function, the mouse must be connected.

Measuring a sample:

1. Right-click to open the menu and click on the *measurement functions menu* (sixth icon from the left).
2. **Calibrate** the microscope.
 - a. Select the sub-menu item *Kalibrierung* and select 10 mm unit.



- b. Place a white paper under the microscope.
- c. Place and focus the microscope slide with a micrometer scale, horizontally – at least 10 mm of the scale must be shown.
- d. Using the mouse, place a point on the scale to the left, and then place the second point at a distance of 10 mm.
- e. Press OK to confirm.



3. Place the holder under the microscope
4. Place the implant between the two roles



5. Right-click the mouse to open the menu and then click on the *measurement functions menu* again.
6. Click on *Linie* and then on *Punkt* to measure from one desired point to another.
7. Press OK
8. Right-click the mouse and select *Speichern* under the sub-menu *Messung*.

Add a horizontal line in the picture:

1. Press the MENU button and use the right arrow button to switch to *Markierung* MENU (fifth icon from the left) or right-click on with the mouse and then click on the icon.
2. Select the sub-menu item *Linienanzeige*.
3. Turn on the horizontal line by selecting *Ein*.
4. If the vertical line is turned on, select the sub-menu *Linientyp* (Line type) and select Horizontal line.
5. Press the MENU button to exit or right-click off the menu



3.5 Getting the pictures

1. Remove the SD card from the microscope
2. Save the pictures from the SD card to another device

If you need an adaptor, contact a supervisor.